TUNGABHADRA COMMAND AREA SOIL TESTING CENTRE, BALLARI

SOIL HEALTH & FERTILITY TEST CERTIFICATE

SAMPLE DATA — FOR PROJECT DEMONSTRATION ONLY (FICTIONAL TEST DATA)

Soil Sample ID:	DEMO-SHC-KA08	**Report Date:**	22-Oct-2026
Location / District:	Siruguppa Taluk, Ballari District, Karnataka	**Analytical Scheme:**	ICAR-NBSS&LUP; Standard Protocol
Soil Classification:	Calcareous Black Soil (Saline-Alkaline Risk)	**Target Agro-Zone:**	Karnataka - Ballari Region

SOIL CHEMICAL AND PHYSICAL PARAMETERS

Parameter Name	Symbol	Observed Value	Unit	Reference Range	Rating / Interpretation
Available Nitrogen	**N**	**192.0**	kg/ha	280.0 - 560.0	Low-Medium (Deficient)
Available Phosphorus	**P**	**29.0**	kg/ha	23.0 - 56.0	Medium (Moderate)
Available Potassium	**K**	**430.0**	kg/ha	140.0 - 330.0	Very High (Abundant)
Soil Reaction (pH)	**pH**	**8.30**	pH scale	6.5 - 7.5	Strongly Alkaline
Electrical Conductivity	**EC**	**1.72**	dS/m	< 1.00	Critical (Slight Salinity Risk)
Organic Carbon	**OC**	**0.40**	%	0.50 - 0.75	Low (Poor)

STANDARD ANALYTICAL READOUT (EXTRACTABLE KEY-VALUE STREAM):

Available Nitrogen (N): 192.0 kg/ha [Low-Medium (Deficient)]

Available Phosphorus (P): 29.0 kg/ha [Medium (Moderate)]

Available Potassium (K): 430.0 kg/ha [Very High (Abundant)]

Soil Reaction (pH): 8.30 [Strongly Alkaline]

Electrical Conductivity (EC): 1.72 dS/m [Critical (Slight Salinity Risk)]

Organic Carbon (OC): 0.40 % [Low (Poor)]

SOIL HEALTH INTERPRETATION & NUTRIENT ADVISORY

Diagnostic Assessment: Intensive command zone black soil showing signs of secondary salinization, calcareous nodules, and alkaline pH, which causes iron and zinc lockup despite surplus potash.

General Nutrient Management Guidance: Install sub-surface or open field drainage trenches to leach soluble salts. Broadcast agricultural gypsum @ 3.5 t/ha. Apply foliar sprays of Ferrous Sulphate (0.5%) and Zinc Sulphate (0.2%) during peak vegetative growth. Favorable for paddy under drainage, sunflower, and cotton.

Soil Analyst: Demonstration Technical Officer

Quality Authorized by: Senior Agricultural Chemist

Status: Fictional Demonstration Data

ACADEMIC DEMONSTRATION DISCLAIMER: *This soil test report contains fictional demonstration data generated strictly for research, user interface validation, and OCR testing in the MannuMitra AI-Based Predictive Farming System. It is not an official laboratory certificate, does not represent physical soil sampling of any real farming parcel, and must not be used as official government documentation or for commercial agricultural transactions.*